

Amy Trent

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EDUCATION

University of British Columbia

Sept. 2025 – Present

Bachelor of Applied Science | Mechanical and Mechatronics Engineering | 3.9/4.33 GPA

Kelowna, BC

- **Completed Courses:** [Introductory Physics I](#), [Atomic and Molecular Chemistry](#), [Computer Programming I](#), [Differential Calculus](#), [Introductory Physics II](#), [Physical and Organic Chemistry](#), [Microeconomics](#), [Integral Calculus](#), [Engineering Drawing and CAD/CAM](#), and [Linear Algebra](#).
- **In-Progress Courses:** [Statics](#), [Materials Science I](#), [Fundamentals of Sustainable Engineering Design](#), [Calculus III](#), [Instrumentation and Data Analysis](#), and [Thermodynamics](#).

EXPERIENCE

Math and Chemistry Tutor

Sept. 2024 – June 2026

- Tutored a handful of different mathematic levels, ranging from the Grade 5 fundamentals to Grade 11 Pre-Calculus, including quadratic functions, polynomial factoring, rational expressions, linear and quadratic inequalities, radical operations, and trigonometry involving non-right triangles.
- Held a drop-in tutoring session for students struggling with Grade 11 Chemistry content, including stoichiometry, Lewis structures, and molarity calculations.

PROJECTS

Dog Cookie Dispenser

Feb. 2021 – April 2021

- Designed a 3D printed dog cookie dispenser that played two similar sounds to test whether my dog could recognize which one signified an incoming cookie.
- Used 3D printing, a rack and pinion system, sensors, a speaker, and C running on an Arduino Nano.

Dragon Head

Sept. 2023 – Jan. 2024

- Designed a 3D printed dragon head with proximity sensors that track where a person is in front of it, causing the dragon's LED eyes to follow the motion, a blue LED to signal a "fire attack", and the ears to move in accordance to its position.
- Used 3D printing, a servo, proximity sensors, LEDs, and C running on an Arudino MEGA2560.

1989 Batmobile

Jan. 2026 – March 2026

- Led a team to design a complete 1989 Batmobile CAD model, designing all parts, assemblies, and mates.
- Used SolidWorks.

Tree Planter

Feb. 2026 – April 2026

- Led a team to create a device that helps tree planters be more efficient with planting, while decreasing the back pain that often results from planting at awkward angles.
- Using 3D printing and SolidWorks.

AI Litter Drone

July 2026 – Current

- Designing a drone that uses a Raspberry Pi to detect litter on the ground, fly to it, pick it up, and dispose of it at a garbage or recycling bin according to its classification of litter.
- Using 3D printing, Raspberry Pi 5 and accessories, machine learning, TBD.

SKILLS

Technical : SolidWorks, Java, C, 3D Printing, Arduino, Raspberry Pi, Machine Learning

Professional : Organization, Communication, Problem-Solving, Leadership, Adaptability, Self-Motivation, Critical-Thinking